

Nabd · نبض

Read where the network goes silent.

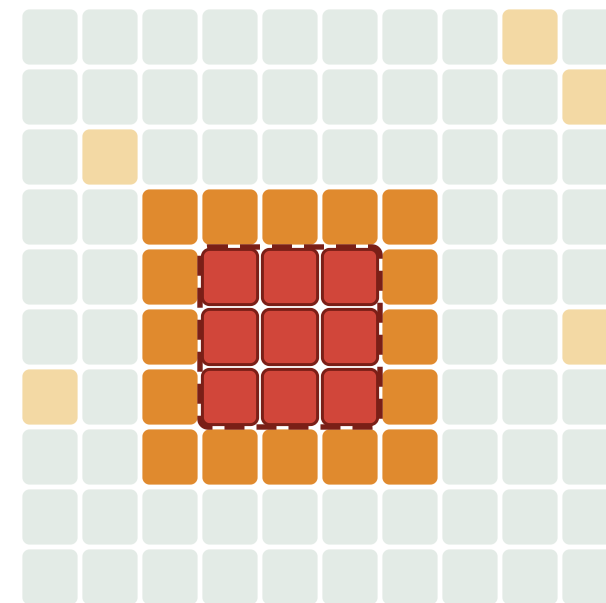
An AI agent that turns coordinated network silence into a live disaster-impact map for command centres — the impact map in the first minute, triage across the first 72 hours.

Theme 6 — Climate Resilience & Environmental Monitoring

Nokia Network-as-Code

Agent layer: LangGraph

Kadir's Team



The first hours decide who is found alive.

53,000+

Kahramanmaraş, Türkiye
earthquakes, February 2023

~3,000

Al Haouz, Morocco
earthquake, September 2023

11,000+

Derna, Libya
dam collapse and flood, September
2023

In each of them, sensors confirmed *something* had happened within seconds.
Which neighbourhoods had lost their people — that map arrived hours late.

2024: Dubai and Oman floods. Same region, a different hazard, the same missing map.

Detection is solved. Impact is not.

SECONDS

Seismometers, satellites and phone-based systems say “**it happened**” almost instantly.

HOURS

“**Which district first?**” is pieced together from fragmented calls, social media and field reports — on a network that is overloaded at exactly that moment.

NABD IS NOT

early warning. Seismometers and existing alert systems own quake detection; we do not compete with them.

NABD IS

the **impact map in the first minute**, and **triage across the first 72 hours** — who is cut off, and where.

Silence is a sensor.

When a whole block of cells goes dark at the same moment, the network already knows where the disaster hit — before a single emergency call gets through.

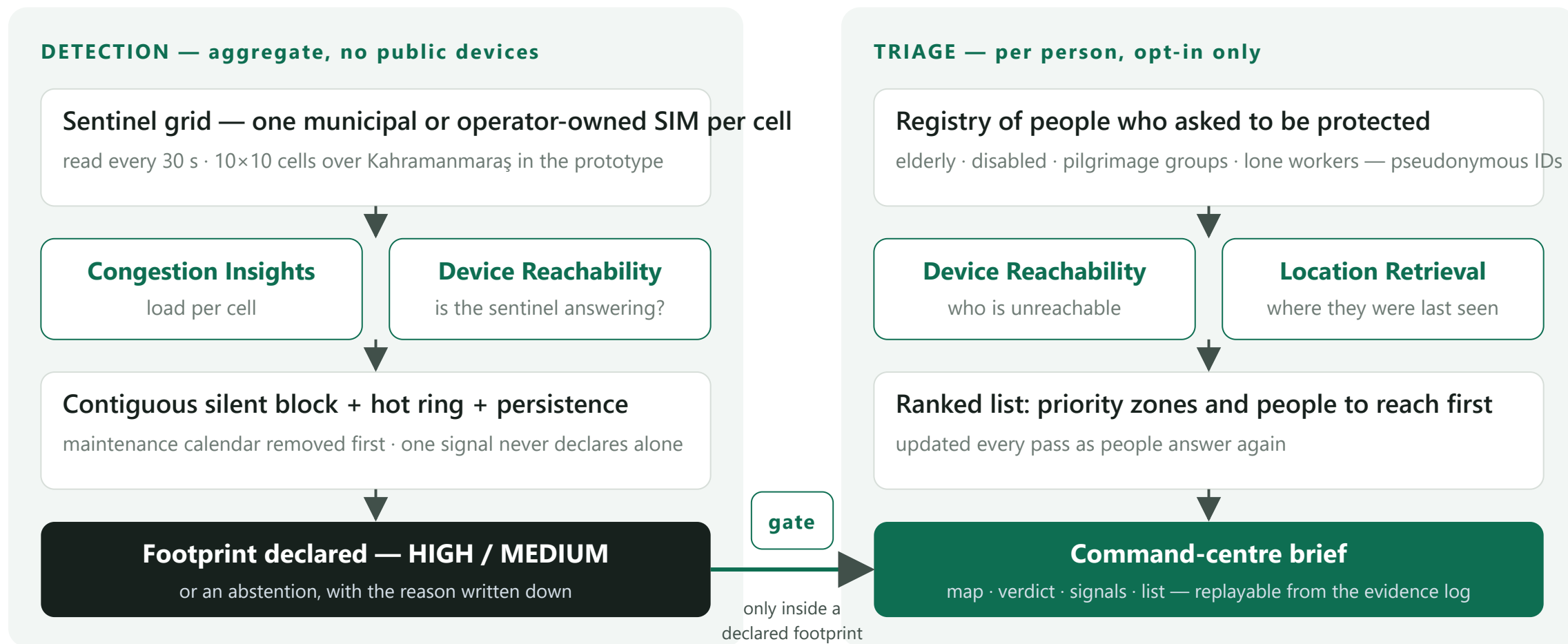
1 An area goes **dark**: its cells stop answering.

2 The ring around it goes **hot**: everyone calls at once.

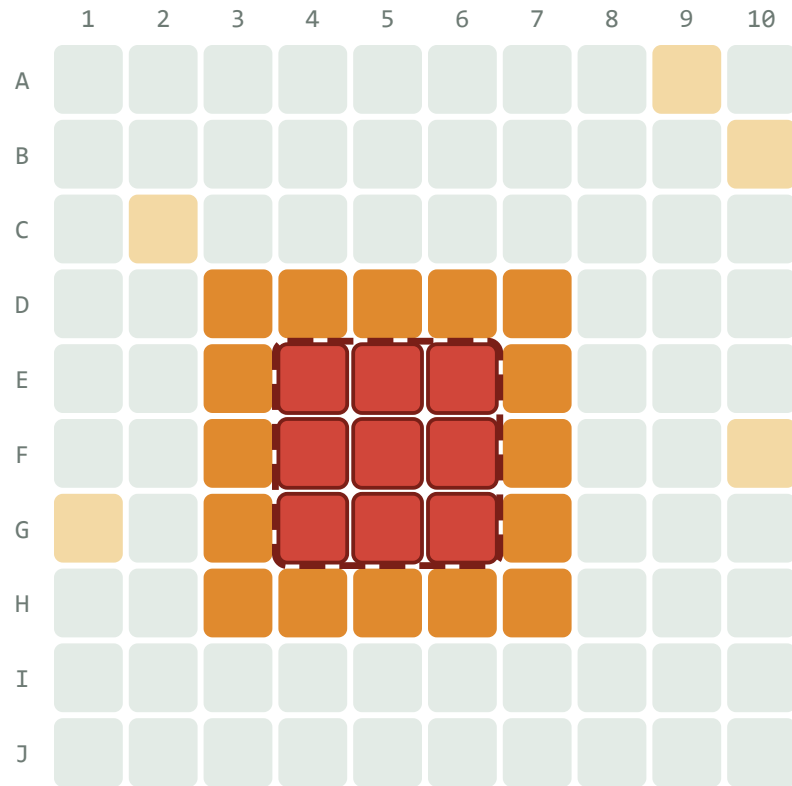
3 That pair, at the same instant, is a signature **no other event produces**.

Disaster-agnostic: earthquake, flood, storm, mass power outage — to the network they are the same thing, an area going silent.

Two layers. Detection is aggregate; triage is consented.



A block goes silent. The ring around it goes hot.



■ silent — sentinel unreachable ■ High congestion ■ Medium
■ Low

The grid is drawn from the evidence record of the declaring pass, not illustrated.

9 cells silent

E4 → G6, about 4.0 km². Their sentinels stop answering — *Device Reachability*.

16 / 16 ring cells at High

Everyone around the block is calling at once — *Congestion Insights*.

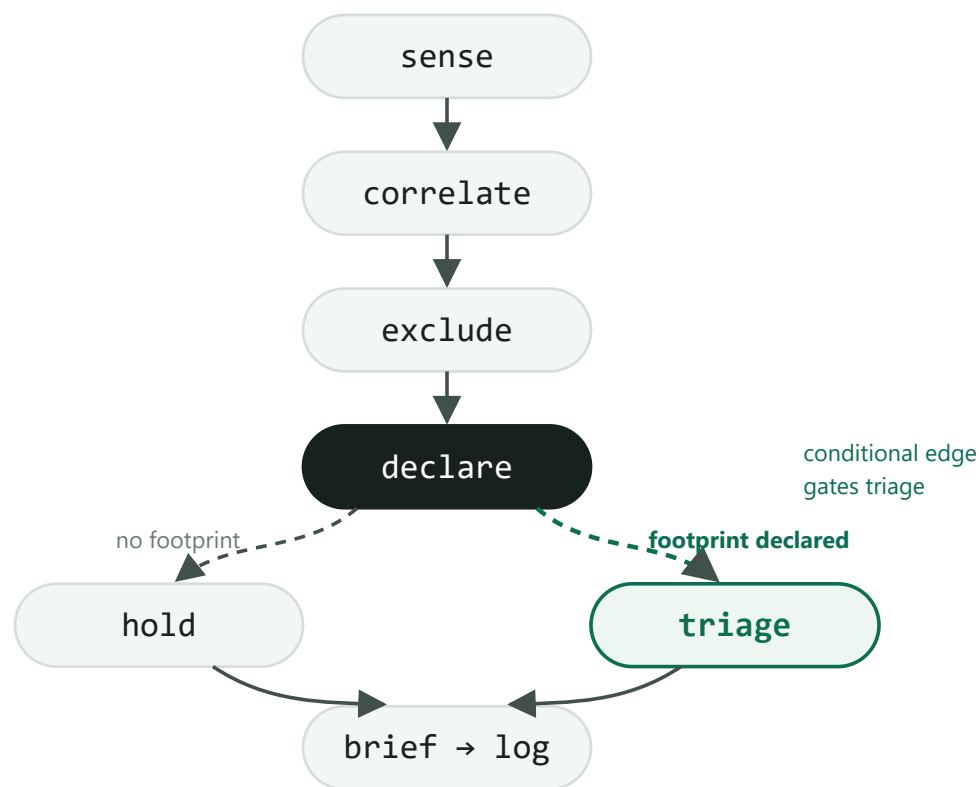
Onset synchronised

All nine went dark inside one 30-second window. A cell fault does not do that.

Persisted a second pass

Candidate at 09:02:30, declared at 09:03:00 — **55 seconds after onset**, confidence **HIGH**.

One signal never declares alone.



Two gates, then corroboration. A footprint needs a contiguous silent block *and* silence that is abnormal *there*, measured against those cells' own history. A gate can veto a declaration; it can never cause one.

Grade the confidence. Corroborations — synchronised onset, hot ring — are a list the arithmetic counts, not a hard-coded pair. Two → HIGH. One → MEDIUM. None → abstain, with the reason written into the log. A new network surface is an append to that list, not a redesign.

Gate the personal layer. The conditional edge means no registry device is ever queried outside a declared footprint — a fuzzed test invariant, not a promise.

Keep the LLM off the control path. Detection is deterministic and replayable; the language model only writes the command-centre sentence.

A working agent, offline and live through one surface.

world · model	10×10 grid, 48-person opt-in registry, quake / cell-fault / peak / chronic-flap events, maintenance calendar
gateway	one CAMARA surface, three backends — simulator, live via the Nokia SDK, and replay of a recorded transcript
detector	correlate → exclude → declare; gates and corroborations as named signals, pure functions, no framework import
triage · brief	reachability and last-seen inside the footprint only; the command-centre sentence
graph · loop	the LangGraph runner and the plain runner — same evidence, byte for byte
shakemap · data	two USGS intensity fields — Kahramanmaraş 2023 and Al Haouz 2023 — reduced and committed with their provenance
llm	the brief's composer, bound to Groq / Gemini / OpenRouter / Ollama with one client — and a guard that rejects any number the model invents
scene · console · parity	six scenes; a self-contained console generated from the evidence; the backend-parity harness

Runs on bare Python

no account, no server, no CDN — the demo cannot fail in the room

Same agent, live

`--backend live` makes the identical three calls through the Nokia SDK and records the raw exchanges

One agent, proven

`python -m nabd.parity` — the agent stack imports no SDK, one function picks the network, every scene replays through the live parsers identically

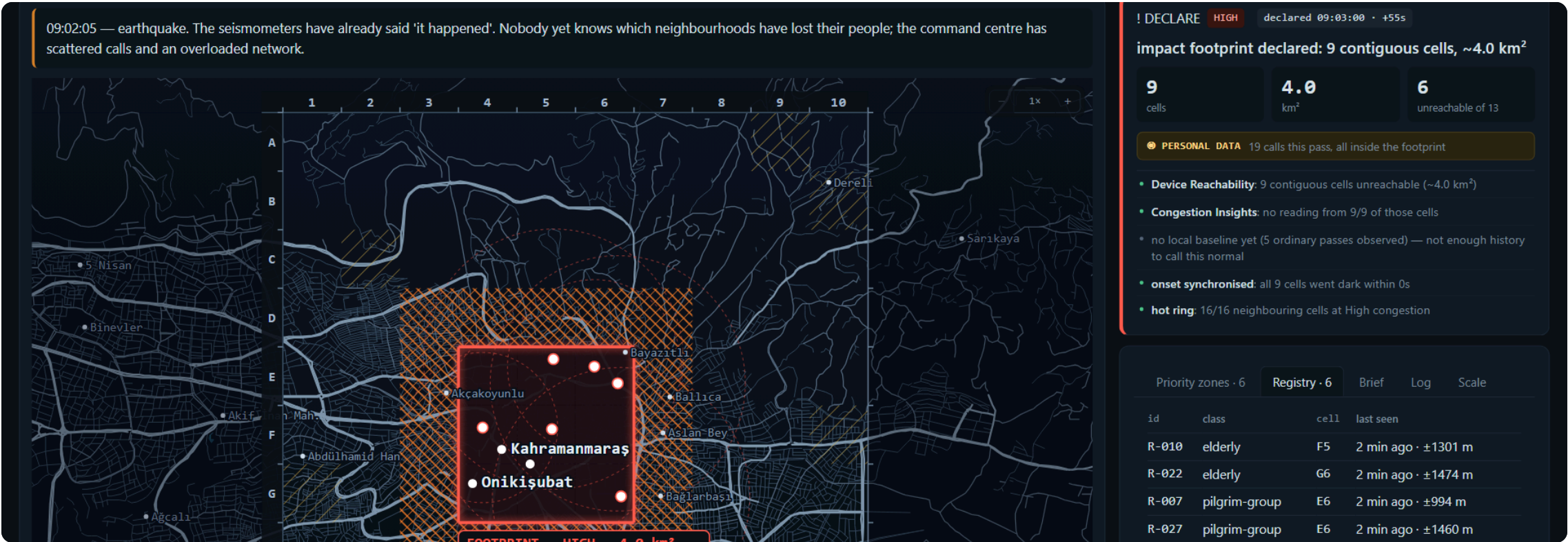
Checked against two real events

Kahramanmaraş drives one scene, Al Haouz another on the same thresholds untouched; nothing named that the measured shaking spared

102 / 102 tests, 13 suites

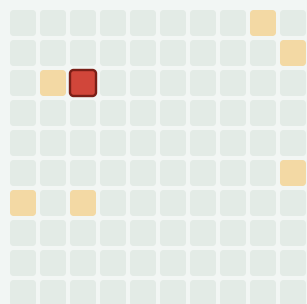
incl. fuzzed invariants, the model's guard, and 0 of 90 ordinary mornings declared under up to 3% random sentinel dropout

Declared 55 seconds after onset. Six people to reach first.



DECLARE · HIGH — 9 cells, ~4.0 km², two gates passed and both corroborations present; 13 opted-in people inside, **6 unreachable**, ranked by need with a last-seen position. Later passes hold the footprint and the list corrects itself 6 → 4 as people answer. Every element on screen is redrawn from the evidence log — the console is a view over the record, never a second source of truth.

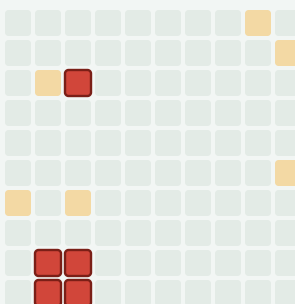
The agent does not cry wolf.



09:01:00 · SINGLE CELL

“single-cell silence at C3: one sentinel unreachable, neighbours normal — consistent with a cell fault, not an impact”

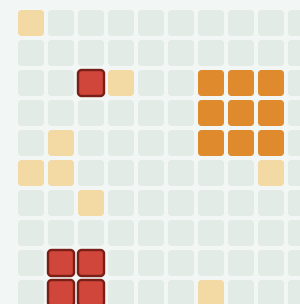
ABSTAIN



09:03:00 · MAINTENANCE

“4 contiguous cells silent, but the block matches maintenance ticket MNT-2214 — expected silence, no alert”

ABSTAIN



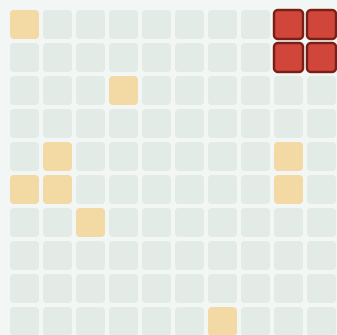
09:05:00 · STADIUM KICK-OFF

“congestion without silence: 9 contiguous cells at High but every sentinel answers — a crowd, not an impact”

ABSTAIN

Noise scene: 21 passes, three abstentions, **zero declarations** — each refusal with its reason in the log. The quiet scene stays quiet. Written reasons are what make a false alarm auditable and a true alarm trusted.

Where silence is normal, silence is not news.

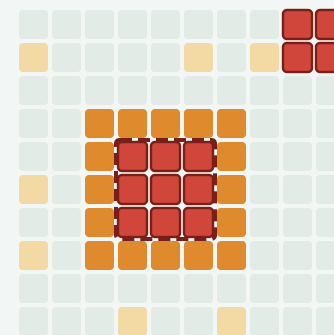


09:05:30 · A BLOCK ON A FAILING BACKHAUL

Four contiguous cells, dark together, no maintenance ticket — **every signal an impact has**. The first three flaps are held for confirmation and come back before it is due. By the fourth, the agent has measured them:

“silence is the local baseline: these 4 cells were already unreachable in 3 of the last 11 passes (27% of the time) — no alert”

ABSTAIN



09:10:30 · SAME RUN, SAME GRID

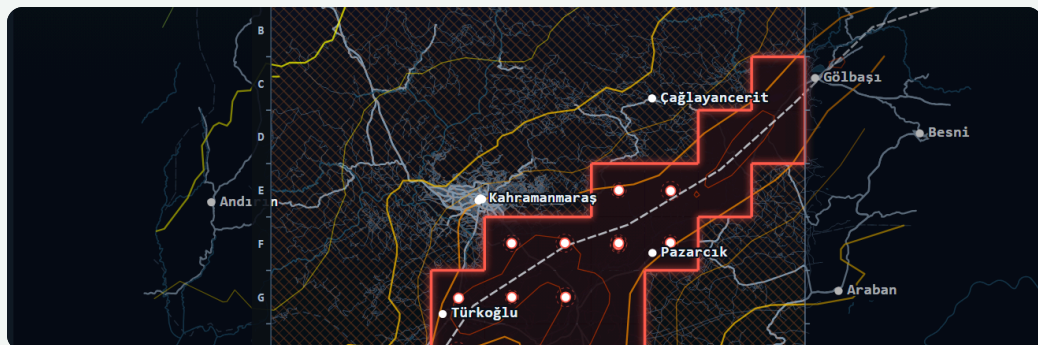
An earthquake hits the centre while the north-eastern block is **still dark**. Two silent blocks on one screen. One alert.

“impact footprint declared: 9 contiguous cells, ~4.0 km²” — HIGH, 30 s after onset, 13 registered people inside

DECLARE

The real question is not whether an outage can be seen — it is whether a **disaster footprint can be told apart from every other reason cells go quiet**. The refusal is measured, not assumed; the declaration in the same run is what shows it is not blindness.

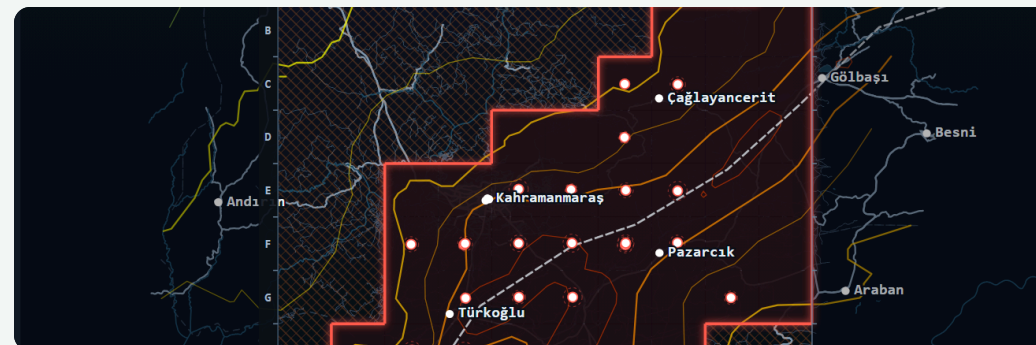
Then we stopped drawing the disaster.



+55 S · THE FIRST MAP

21 contiguous cells, ~2,100 km², HIGH. Every cell it names is one the measured shaking places above the collapse threshold — no invented damage. Two more, an isolated pocket below the three-cell floor, are **deliberately not claimed**.

DECLARE



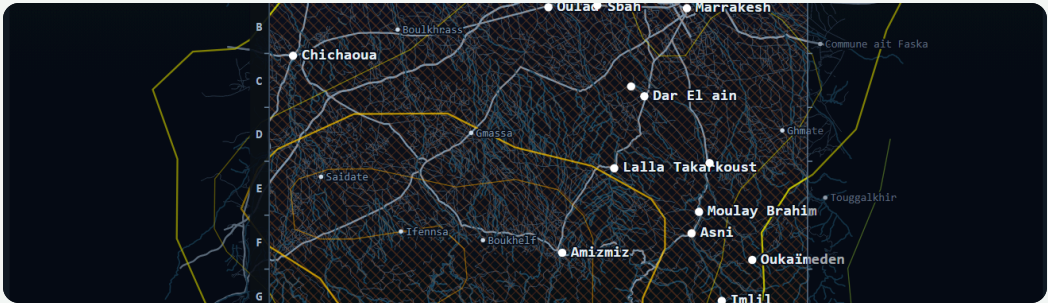
+9 MIN · THE NETWORK KEEPS DYING

63 cells, ~6,300 km², still HIGH. Masts that survived the shaking lost mains power and drained their batteries, worst-shaken first. The pocket has joined. 22 registered people unreachable, medical-dependent first.

UPDATE ×12

The coloured lines are the measured isoseismals and the dashed one is the fault rupture, so the footprint can be checked against the shaking that caused it by eye. Geometry and intensity are the **USGS ShakeMap for the M7.8 Pazarlık earthquake**, 6 Feb 2023 — 262 seismic stations, 1,459 intensity observations. Ours is only the rule turning shaking into silence, calibrated to Turkcell's “**more than half of local base stations inoperative**” — our window ends 63% dark.

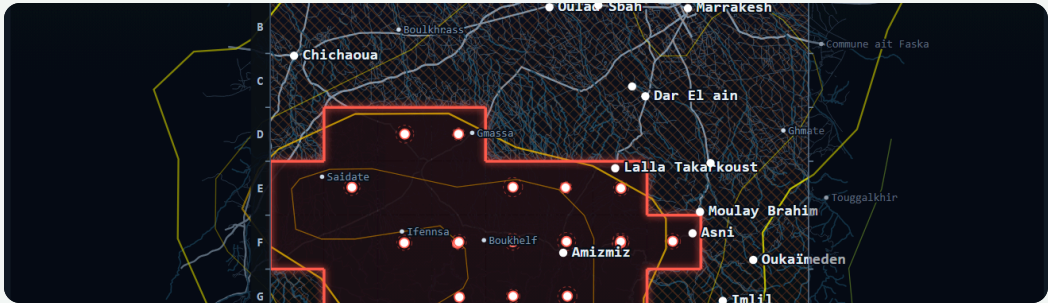
Then we ran it where it was never tuned.



+25 S · THE FIRST VERDICT IS A REFUSAL

Two silent cells at the epicentre, below the three-cell floor. An M6.8 at 19 km does not flatten a block, and the agent says so instead of declaring. It declares at **+235 s, MEDIUM** — the onset genuinely was staggered, and the evidence says why.

ABSTAIN → DECLARE



+9 MIN · ARRIVED THROUGH THE BATTERIES

39 cells, ~3,900 km², still MEDIUM — exactly the set the measured field puts at or above the power threshold. Marrakesh, at intensity 6.4 in the north, answered throughout and is **never named**. 19 registered people unreachable at peak.

UPDATE ×10

	KAHRAMANMARAŞ · 6 FEB 2023	AL HAOUZ · 8 SEP 2023
ShakeMap	us6000jllz v19 · M7.8 · 262 stations	us7000kufc v14 · M6.8 · 3 stations
Thresholds	calibrated to Turkcell's "more than half inoperative"	the same numbers, untouched — asserted by a test
First verdict	DECLARE at +55 s · 21 cells · HIGH	ABSTAIN at +25 s · below the floor
Declaration	+55 s · HIGH · onset synchronised	+235 s · MEDIUM · onset staggered
A city that answered	—	Marrakesh — shaken, on every television, never named

Remove the network APIs and nothing works. The network is the sensor.

API	ROLE	PROTOTYPE
Congestion Insights	detection — load per sentinel cell, the hot ring	live path
Device Reachability Status	detection (sentinels) and triage (registry)	live path
Location Retrieval	triage — last-seen position of the unreachable	live path
Geofencing Subscriptions	boundary events as the footprint grows or shrinks	roadmap
Quality on Demand	a priority channel for responders inside the footprint	roadmap

WHY CAMARA, NOT A CONTROL ROOM

An operator's NOC sees its own cells go down — from inside one operator.

Responders — civil defence, the Red Crescent, municipalities, hospitals — sit **outside**, and cannot log into three operators in the middle of a disaster.

Open Gateway is the **one standard door**: operator-independent, and it unifies every national operator behind a single interface. Stable since 2025, available on Network-as-Code today.

Live where it can be. Simulated where it must be. Private by design.

LIVE VS SIMULATED — AND THE SEAM BETWEEN THEM

The sandbox cannot stage a disaster, so the two claims are separated: the **live platform proves the integration**; the **simulator proves the scenario**.

The risk is the seam between them, so `nabd.parity` measures it: no SDK in the agent stack, **one function** picks the network, every scene **replayed through the live parsers** identically. The live check reports **pending** until a key runs it — never pass.

PRIVACY

Detection reads only **sentinel devices the city or operator owns** — never the public.

Its output is **area-level**: cells and a footprint, not people. No cameras, no message content, no tracking.

CONSENT

Triage touches an individual only if they **opted in** to be protected.

The agency holds the registry; the operator honours it. IDs are pseudonymous, and **no location call is ever made outside a declared footprint**.

THE MODEL, WATCHED it phrases the brief and nothing else; a guard reads its sentence back against the facts — **an invented number throws the sentence away**. · **FALSE ALARMS, MEASURED** **0 of 90** ordinary mornings declared under 1–3% random sentinel dropout; the edge near 5% is a written deployment requirement.

One engine, many hazards. Minutes, not hours.

IMPACT

Earthquake, flood, storm, mass outage — to the network each is an area going dark, so one engine covers them all.

The first dispatch decision moves from **hours to minutes**, and the most vulnerable are reached first.

SCALE

A sentinel grid scales by **cell count**, not population: **7,836 SIMs watch all of Türkiye**, 4,466 all of Morocco — 188 k and 107 k CAMARA calls an hour at a five-minute watch that tightens to 30 s where something happens.

The sentinels already exist: fixed SIMs in meters, traffic lights, pumps and site monitors, opted in by their owners — a procurement, not a behaviour change.

BUSINESS

Buyers: civil-defence agencies (AFAD), Red Crescent societies, municipalities, hospital networks.

Seller: the operator, as an Open Gateway **impact-feed product** — a standing subscription for the grid, per-event triage on top.

ROADMAP Population Density Data as the detection upgrade once published on NaC · Geofencing for boundary events · QoD priority for responders inside the footprint · multi-operator fusion

The network already knows. Nabd makes it say so — in the first minute.

NEXT

Live contract run on Network-as-Code devices ·
a third real event, a flood, on the same
thresholds · pilot with one civil-defence agency
on one city's sentinel grid

TEAM

Kadir Can Yıldırım — Kadir's Team, solo builder,
Türkiye. Code, evidence logs and the command-
centre console are in the submission links.

See you in Doha — MWC Doha, November 2026.

